

Chemistry - Question 121

Original paper page 48; original question and option regions

121. In which of the following, the product of uncertainty in velocity and uncertainty in position of a micro particle of mass 'm' is not less than

1) $h \times \frac{3\pi}{m}$

2) $\frac{h}{3\pi} \times m$

3) $\frac{h}{4\pi} \times \frac{1}{m}$

4) $\frac{h}{4\pi} \times m$

Key: 3

Sol: $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

not less than $\frac{h}{4\pi} \times \frac{1}{m}$.

Chemistry - Question 122

Original paper page 48; original question and option regions

122. An element has $[Ar]3d^1$ configuration in its +2 oxidation state. Its position in the periodic table is

- 1) period -3, group-3 2) period-3, group-7 3) period-4, group-3 4) period-3, group-9

Key: 3

Sol:Ground state configuration $(Ar)4s^23d^1$

Scandium- Sc

Period number = ultimatate shell number

$\therefore$ Period number = 4

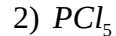
Valency electrons = 3

Group number = 3

Chemistry - Question 123

Original paper page 48; original question and option regions

123. In which of the following molecules all bond lengths are not equal?



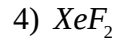
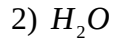
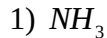
Key: 2

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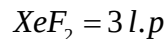
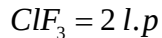
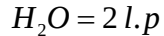
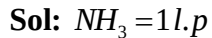
Chemistry - Question 124

Original paper page 49; original question and option regions

124. In which of the following molecules maximum number of lone pairs is present on the central atom?



Key: 4



Chemistry - Question 125

Original paper page 49; original question and option regions

125. Which one of the following is the kinetic energy of a gaseous mixture containing 3 g of hydrogen and 80 g oxygen at temperature T(K)?

1) 3 RT

2) 6 RT

3) 4 RT

4) 8 RT

Key: 2

Sol: $K.E = \frac{3}{2} nRT$

$$= \frac{3}{2} (n_{O_2} + n_{H_2}) RT$$

$$= \frac{3}{2} \left(\frac{80}{32} + \frac{3}{2} \right) RT$$

$$= \frac{3}{2} \times 4RT = 6RT .$$

Chemistry - Question 126

Original paper page 49; original question and option regions

126. A, B, C and D are four different gases with critical temperatures 304.1, 154.3, 405.5 and 126.0 K respectively. While cooling the gas which gets liquified first is

1) B

2) A

3) D

4) C

Key: 4

Sol: Critical temperature $\propto$ liquification

Chemistry - Question 127

Original paper page 49; original question and option regions

127. 40 ml of x M $KMnO_4$ solution is required to react completely with 200 ml of 0.02 M oxalic acid solution in acidic medium. The value of x is

1) 0.04

2) 0.01

3) 0.03

4) 0.02

Key: 1

Sol: $KMnO_4$

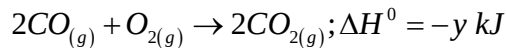
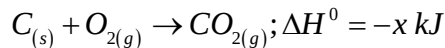
xM

$H_2C_2O_4$

0.02 M

Chemistry - Question 128

Original paper page 50; original question and option regions

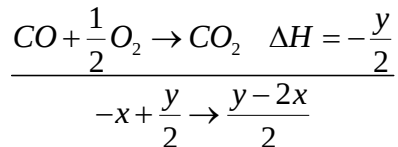
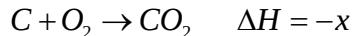
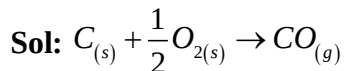
128. Given that**The enthalpy of formation of CO will be**

1) $\frac{y-2x}{3}$

2) $\frac{y-2x}{2}$

3) $\frac{2x-y}{2}$

4) $\frac{x-y}{2}$

Key: 2

Chemistry - Question 129

Original paper page 50; original question and option regions

129. At 400 K, in a 1.0 L vessel N_2O_4 is allowed to attain equilibrium $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. At equilibrium the total pressure is 600 mm Hg, when 20 % of N_2O_4 is dissociated. The K_p value for the reaction is

1) 50

2) 100

3) 150

4) 200

Key: 2

Sol: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

1	0
1-0.2	2×0.2
0.8	0.4

Total no. of mole= 1.2

$$K_p = \frac{n_{NO_2}^2}{n_{N_2O_4}} \left(\frac{P}{\sum n} \right)^{\Delta n}$$

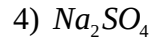
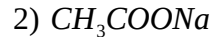
$$= \frac{(0.4)^2}{0.8} \left(\frac{600}{1.2} \right)^1$$

$$K_p = 100$$

Chemistry - Question 130

Original paper page 50; original question and option regions

130. In which of the following salts only cationic hydrolysis is involved?

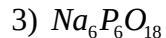
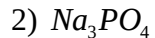
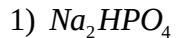


Key: 3

Chemistry - Question 131

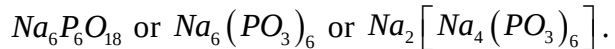
Original paper page 51; original question and option regions

131. Calgon is



Key: 3

Sol: Sodium hexa meta phosphate is called calgon



Chemistry - Question 132

Original paper page 51; original question and option regions

132. Consider the following statements

I) CS^+ ion is more highly hydrated than other alkali metal ions

II) Among the alkali metals, only lithium forms a stable nitride by direct combination with nitrogen

III) Among alkali metals Li, Na, K, Rb, the metal, Rb has the highest melting point

IV) Among alkali metals Li, Na, K, Rb only Li forms peroxide when heated with oxygen

The correct statement is

1) I

2) II

3) III

4) IV

Key: 2

Sol: $6Li + N_2 \rightarrow 2Li_3N$.

Chemistry - Question 133

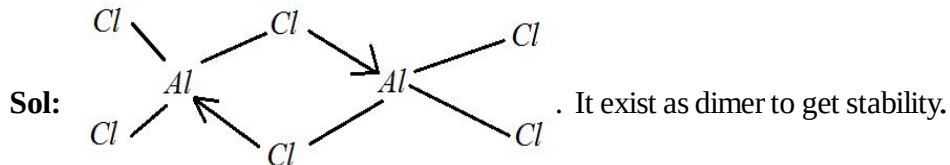
Original paper page 51; original question and option regions

133. Assertion (A): $AlCl_3$ exists as a dimer through halogen bridged bonds.

Reason (R): $AlCl_3$ gets stability by accepting electrons from the bridged halogen.

- 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- 2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- 3) (A) is true, but (R) is not true
- 4) (A) is not true, but (R) is true.

Key: 1



Chemistry - Question 134

Original paper page 51; original question and option regions

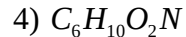
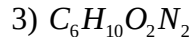
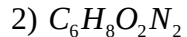
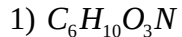
**134. Which of the following causes
“Blue baby syndrome”**

- 1) High concentration of lead in drinking water
- 2) High concentration of sulphates in drinking water
- 3) High concentration of nitrates in drinking water
- 4) High concentration of copper in drinking water

Chemistry - Question 135

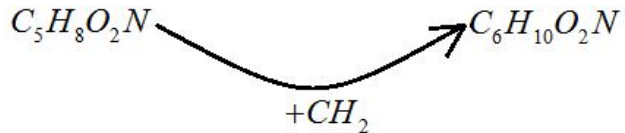
Original paper page 52; original question and option regions

135. Which of the following belongs to the homologous series of $C_5H_8O_2N$?



Key: 4

Sol: Homologous series means difference is CH_2 group



Chemistry - Question 136

Original paper page 52; original question and option regions

136. In Dumas method, 0.3 g of an organic compound gave 45 mL of nitrogen at STP. The percentage of nitrogen is

1) 16.9

2) 18.7

3) 23.2

4) 29.6

Key: 2

$$\text{Sol: } \%N = \frac{\text{Volume of } N_2 \text{ at STP (ml)}}{\text{wt. of organic compound}} \times \frac{28}{22400} \times 100$$

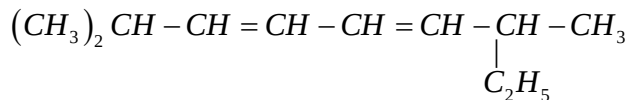
$$= \frac{45}{0.3} \times \frac{28}{22400} \times 100$$

$$= 18.75 \text{ ml}$$

Chemistry - Question 137

Original paper page 52; original question and option regions

137. The IUPAC name of



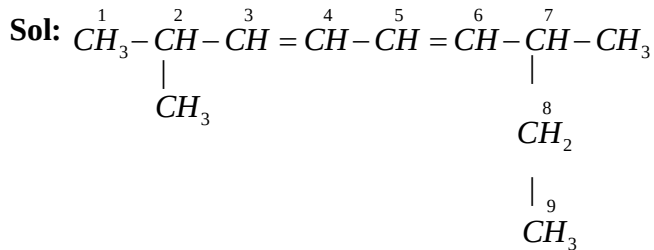
1) 2, 7- dimethyl -3, 5 - nonadiene

2) 2, 7- dimethyl -2- ethylheptadiene

3) 2- methyl -7- ethyl -3, 5- octadiene

4) 1, 1- dimethyl- 6 - ethyl -2, 4- heptadiene

Key: 1



2, 7 dimethyl 3, 5- nonadiene

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Chemistry - Question 138

Original paper page 52; original question and option regions

138. Match the following

List-I

(magnetic property)

A) Ferromagnetic

B) Anti ferro magnetic 2) CrO_2

C) Ferri magnetic

D) Para magnetic

List - II

(substance)

1) O_2

3) MnO

4) Fe_3O_4

5) C_6H_6

Chemistry - Question 139

Original paper page 53; original question and option regions

139. The vapour pressure of pure benzene and toluene are 160 and 60 mmHg respectively. The mole fraction of benzene in vapour phase in contact with equimolar solution of benzene and toluene is

1) 0.073

2) 0.027

3) 0.27

4) 0.73

Key: 4

Sol: $P_A = Y_A \cdot P$

$$Y_A = \frac{P_A^0 X_A}{P_A^0 X_A + P_B^0 X_B} = \frac{160 \times X_A}{160 X_A + 60 X_A}$$

$$= 0.66$$

$$X_A = \frac{n_A}{n_A + n_B} = \frac{1}{2}$$

$$X_A = \frac{160 \times 0.5}{160 \times 0.5 + 60 \times 0.5} = \frac{160}{220} = 0.7272$$

$$= 0.73$$

Chemistry - Question 140

Original paper page 53; original question and option regions

140. 6g of a non volatile, non electrolyte X dissolved in 100 g of water freezes at -0.93°C . The molar mass of X in g mol^{-1} is $(K_f \text{ of } \text{H}_2\text{O} = 1.86 \text{ K kg mol}^{-1})$

1) 60

2) 140

3) 180

4) 120

Key: 4

Sol: $\Delta T_f = K_f \times m$

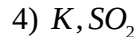
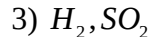
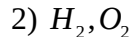
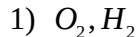
$$0 - (-0.93) = 1.86 \times \frac{6}{M} \times \frac{1000}{100}$$

$$M = 120 .$$

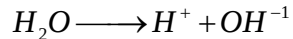
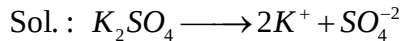
Chemistry - Question 141

Original paper page 53; original question and option regions

141. The products obtained at the cathode and anode respectively during the electrolysis of aqueous K_2SO_4 solution using platinum electrodes are



Key : 2



Chemistry - Question 142

Original paper page 54; original question and option regions

142. The slope of the graph drawn between $\ln k$ and $1/T$ as per Arrhenius equation gives the value (R = gas constant, E_a = Activation energy)

1) $\frac{R}{E_a}$

2) $\frac{E_a}{R}$

3) $\frac{-E_a}{R}$

4) $\frac{-R}{E_a}$

Key : 3

Sol. : $K = Ae^{-E_a/RT}$

$$\log_{10} K = \log A - \frac{E_a}{RT}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$\ln K = \left(\frac{-E_a}{R} \right) \frac{1}{T} + \ln A$$

$$y = mx + c$$

$$\text{slope} = \frac{-E_a}{R}$$

Chemistry - Question 143

Original paper page 54; original question and option regions

143. Which is not the correct statement in respect of chemisorption?

- | | |
|-------------------------------|--------------------------------|
| 1) Highly specific adsorption | 2) Irreversible adsorption |
| 3) Multilayered adsorption | 4) High enthalpy of adsorption |

Key : 3

Sol. : Conceptual

Chemistry - Question 144

Original paper page 54; original question and option regions

144. Which of the following is carbonate ore?

1) Cuprite

2) Siderite

3) Zincite

4) Bauxite

Key : 2

Sol. : Cuprite $\rightarrow Cu_2O$

Siderite $\rightarrow FeCO_3$

Zincite $\rightarrow ZnO$

Bauxite $\rightarrow Al_2O_3 \cdot 2H_2O$

Chemistry - Question 145

Original paper page 54; original question and option regions

145. Which one of the following statements is not correct?

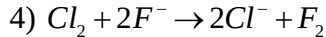
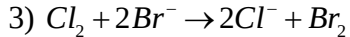
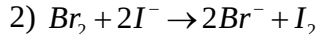
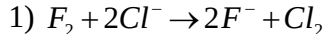
- 1) O_3 is used as germicide
- 2) In O_3 , O-O bond length is identical with that of molecular oxygen
- 3) O_3 is an oxidising agent
- 4) The shape of O_3 molecule is angular.

Key : 2

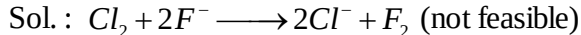
Chemistry - Question 146

Original paper page 55; original question and option regions

146. Which of the following reactions **does not** take place



Key : 4



This reaction does not take place because fluorine is strong oxidising agent (under goes reduction)

Oxidising nature : $F_2 > Cl_2 > Br_2 > I_2$

Chemistry - Question 147

Original paper page 55; original question and option regions

147. Which of the following statements regarding sulphur is not correct?

- 1) At about 1000K, it mainly consists of S_2 molecules
- 2) The oxidation state of sulphur is never less than +4 in its compounds
- 3) S_2 molecule is paramagnetic
- 4) Rhombic sulphur is readily soluble in CS_2

Key : 2

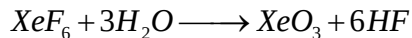
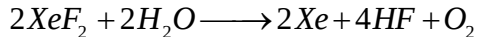
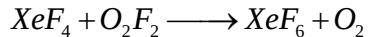
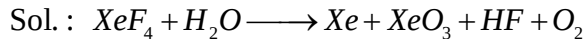
Sol. : It shows less than +4 oxidation

Ex: $H_2S \Rightarrow S = -2$

148. Which of the following reactions does not involve, liberation of oxygen?



Key : 4



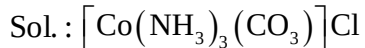
Chemistry - Question 149

Original paper page 55; original question and option regions

149. Select the correct IUPAC name of $[Co(NH_3)_5(CO_3)]Cl$

- 1) Penta ammonia carbonate cobalt (III) chloride
- 2) Pentammine carbonate cobalt chloride
- 3) Pentammine carbonato cobalt (III) chloride
- 4) Cobalt (III) pentammine carbonate chloride.

Key : 3



Penta ammine carbonate cobalt (III) chloride.

$Co \longrightarrow$ oxidation no. is 3

Chemistry - Question 150

Original paper page 56; original question and option regions

150. Which of the following characteristics of the transition metals is associated with their catalytic activity?

1) Colour of hydrated ions

2) Diamagnetic behaviour

3) Paramagnetic behaviour

4) Variable oxidation states

Key : 4

Sol. : Due to variable oxidation states

They exhibit different colour.

Chemistry - Question 151

Original paper page 56; original question and option regions

151 Observe the following polymers

(A) PHBV; (B) Nylon 2-nylon 6; (C) Glyptal; (D) Bakelite

1) (C)

2) (A), (B)

3) (D)

4) (C), (D)

Key : 2

Sol. : $\left. \begin{array}{l} \text{PHBV} \\ \text{Nylon-2 - nylon-6} \end{array} \right] \rightarrow \text{Biodegradable polymers}$

$\left. \begin{array}{l} \text{Glyptal} \\ \text{Bakelite} \end{array} \right] \rightarrow \text{Non-Biodegradable polymers}$

Chemistry - Question 152

Original paper page 56; original question and option regions

152. Observe the following statements

i : Sucrose has glycosidic linkage; ii : Cellulose is present in both plants and animals

iii : Lactose contains D-galactose and D-glucose units

The correct statements are

1) (i), (ii), (iii)

2) (i), (ii)

3) (ii), (iii)

4) (i), (iii)

Key : 4

Sol. : Cellulose present in only plants but not in animals.

Glucose has glycosidic linkage between

C_1 of α -D glucose and C_2 of β -D-Fructose

Lactose is composed of

β -D-Galactose and β -D-glucose

Chemistry - Question 153

Original paper page 56; original question and option regions

153. Identify the antioxidant used in foods

1) Aspartame

2) Sodium benzoate

3) Ortho-sulpho benzimide

4) Butylated hydroxy toluene

Key : 4

Sol. : Aspartame - Artificial sweetening agent

Sodium Benzoate - Food preservative

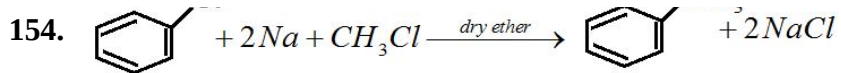
Butylated hydroxy toluene (BHT) - Anti oxidant

Cl

CH₃

Chemistry - Question 154

Original paper page 56; original question and option regions



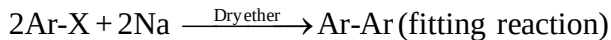
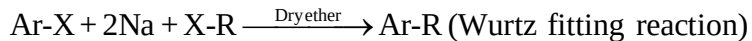
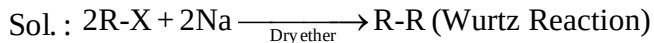
This reaction is known as

1) Wurtz-Fitting reaction 2) Wurtz reaction

3) Fitting reaction

4) Friedel-crafts reaction

Key : 1



Chemistry - Question 155

Original paper page 57; original question and option regions

155. What is Z in the following sequence of reactions?



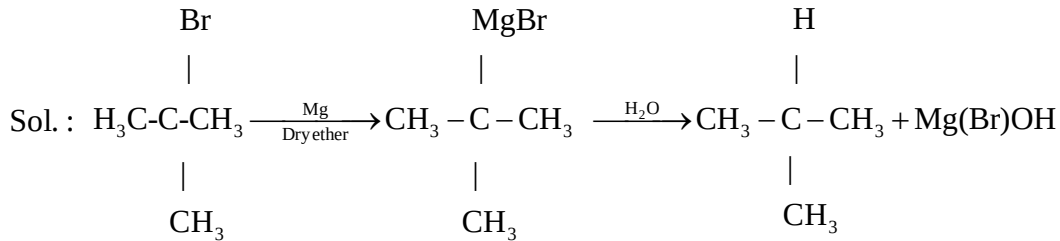
1) propane

2) 2-methyl propene

3) 2-methyl propane

4) 2-methyl butane

Key : 3

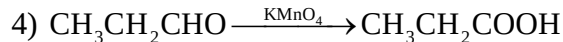
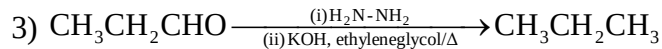
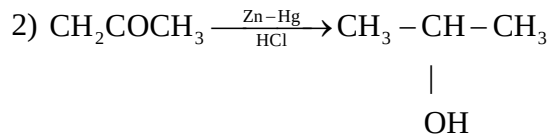


(2 methyl propane)

Chemistry - Question 156

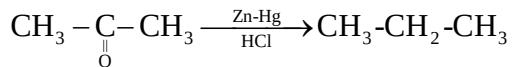
Original paper page 57; original question and option regions

156. In which of the following reactions the product is not correct?

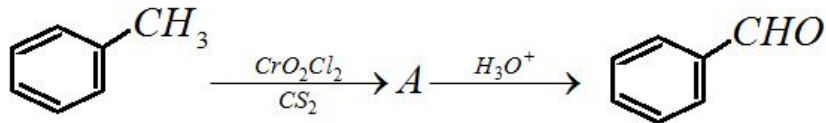


Key : 2

Sol. : Zn-Hg/HCl $\longrightarrow$ Clemanson Reagent



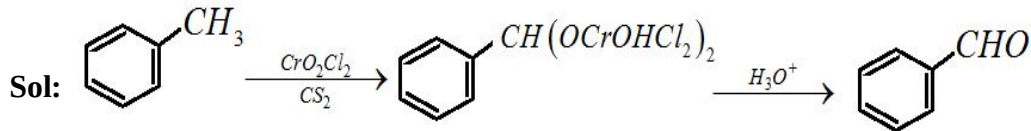
157. Identify the name of the following reaction



- 1) Gatterman - Koch reaction
- 3) Stephen reaction

- 2) Gatterman reaction
- 4) Etard reaction

Key : 4

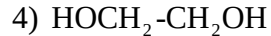
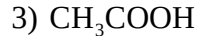
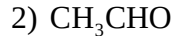
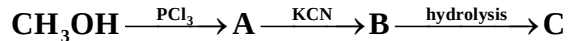


Etard reaction.

Chemistry - Question 158

Original paper page 57; original question and option regions

158. What is C in the following sequence of reactions?



Key : 3

Chemistry - Question 159

Original paper page 58; original question and option regions

159. The order of basic strength of the following in aqueous solution is

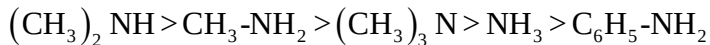
(1) $\text{C}_6\text{H}_5\text{NH}_2$; (2) $(\text{CH}_3)_3\text{N}$; (3) NH_3 ; (4) CH_3NH_2 ; (5) $(\text{CH}_3)_2\text{NH}$

1) $4 > 1 > 5 > 3 > 2$ 2) $2 > 5 > 4 > 3 > 1$ 3) $5 > 4 > 2 > 3 > 1$ 4) $4 > 3 > 5 > 2 > 1$

Key : 3

Sol. : Basicity of amines

Aliphatic $> \text{NH}_3 > \text{Aromatic}$



Chemistry - Question 160

Original paper page 58; original question and option regions

160. Yellow dye can be prepared by a coupling reaction of benzene diazonium chloride in acid medium with X. Identify X from the following

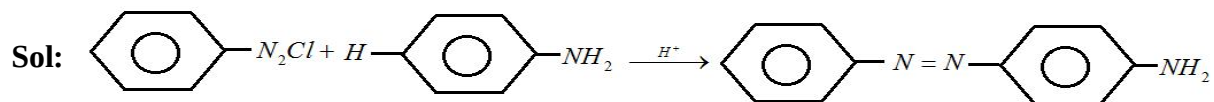
1) Aniline

2) Phenol

3) Cumene

4) Benzene

Key : 1



X= Aniline ($C_6H_5NH_2$).

(Yellow dye)
